ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 6

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 70

Marking guideline for the Academic Central Grade 6 Natural Sciences and Technology paper 1 · practice a.

Written by Academic Central for practice, following the CAPS curriculum for Grade 6. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples. Do not take marks off for spelling if the meaning is clear.

QUESTION 1

1 1.1 B ✓ 1.2 C ✓ 1.3 B ✓ 1.4 A ✓ 1.5 B ✓ 1.6 C ✓ 1.7 B ✓ 1.8 B ✓ 1.9 B ✓ 1.10 C ✓ (10)

[10]

QUESTION 2

2 2.1 (Electrical) insulator ✓ 2.2 (Electric) circuit ✓ 2.3 Battery ✓ 2.4 Illegal connection ✓ 2.5 Orbit ✓ 2.6 A year (365¼ days) ✓ 2.7 Moon (natural satellite) ✓ 2.8 Astronaut ✓ (8)

[8]

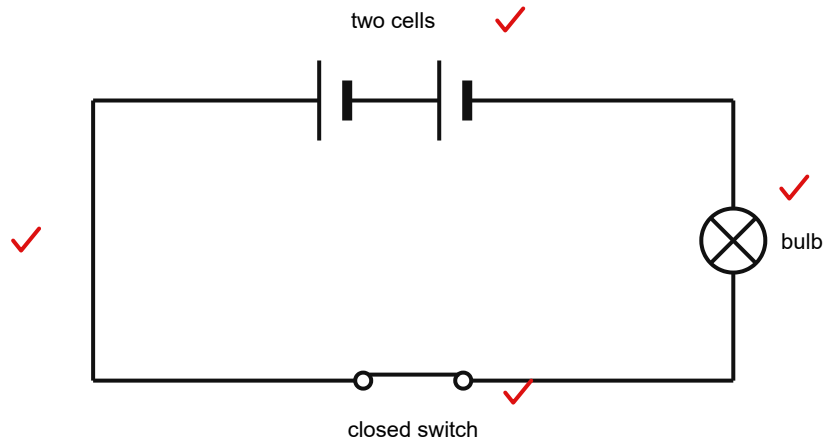
QUESTION 3

3.1.1 A cell ✓ B switch ✓ C bulb (lamp) ✓ (3)

3.1.2 No ✓. The switch is open, so the circuit is not complete (it is broken) ✓ (2)

3.1.3 It gives the energy that pushes the electricity around the circuit ✓ (1)

3.2



Two cells ✓; closed switch ✓; bulb ✓; complete circuit with all parts joined ✓ (4)

3.3.1 The steel nail ✓, the aluminium foil ✓ and the coin ✓ (3)

3.3.2 (Electrical) conductors ✓ (1)

3.3.3 Plastic is an insulator ✓, so it stops electricity from flowing into a person who touches the cable (prevents shocks and short circuits) ✓ (2)

3.4 Add another cell (use more cells) ✓ (1)

3.5 Any ONE: the bulb is broken (blown) ✓; the cell is flat; a wire is loose or broken; the bulb is not screwed in properly (1)

[18]

QUESTION 4

4.1 B → D → A → C → E ✓✓✓ (all correct: 3 marks; take off 1 mark for each letter in the wrong place) (3)

4.2 Any TWO: smoke and gases pollute the air ✓; gases add to global warming (climate change) ✓; acid rain; coal mines damage the land; power stations use a lot of water; ash dumps (2)

4.3 Any TWO: the Sun (solar power) ✓; wind ✓; moving water (hydro-electric power); waves (2)

4.4 Any TWO: people, especially children, can be shocked or killed ✓; they cause fires ✓; they overload the system and cause power cuts; they damage appliances (2)

4.5 Any THREE: switch off lights when leaving a room ✓; switch appliances off at the wall ✓; boil only as much water as you need ✓; use energy-saving (LED) bulbs; dry clothes in the sun; use a solar geyser or switch the geyser off when not needed; close the fridge door quickly (3)

[12]

QUESTION 5

5.1.1 Earth ✓ (1)

5.1.2 Jupiter ✓, number 5 ✓ (2)

5.1.3 Saturn ✓ (1)

5.1.4 They are very far from the Sun, so they get very little heat ✓ (1)

5.1.5 Mercury, Venus, Earth and Mars ✓✓ (all four: 2 marks; two or three: 1 mark) (2)

5.2 Rotation: the Earth spins (turns) on its axis ✓; it takes 24 hours (one day) ✓
Revolution: the Earth moves around the Sun in its orbit ✓; it takes 365¼ days (one year) ✓ (4)

5.3.1 A full moon ✓ C new moon ✓ (2)

5.3.2 The Moon does not make its own light; it reflects light from the Sun ✓. As the Moon moves around the Earth, we see different amounts of its lit half ✓ (2)

5.3.3 About 27 to 29 days (about a month) ✓ (1)

5.4 Any ONE: the Moon has craters and mountains ✓; Jupiter has moons; Saturn has rings; there are millions of stars and other galaxies; the planets move around the Sun (1)

5.5 Any TWO: there is no air (oxygen) to breathe ✓; it is far too hot or too cold ✓; there is no liquid water to drink; no food grows there; dangerous rays from the Sun (2)

5.6 Any ONE: Mars is very far away and the journey is too long and dangerous ✓; robots do not need air, food or water; it is cheaper; robots do not have to come back (1)

5.7 Any TWO: air (oxygen) ✓; water ✓; food; a spacesuit; medicine (2)

[22]

TOTAL: 70

END OF MARKING GUIDELINE